

- $\text{speed} = \text{ceil}(\text{SpeedMax} * (\text{percent} * 0.01))$

For example: If the SpeedMax attribute is 42 (42 speed fan) and PercentSetting is changed to 25, then [SpeedSetting](#) and [SpeedCurrent](#) become 11 (rounding up 10.5).

4.4.6.4. PercentCurrent Attribute

This attribute SHALL indicate the actual currently operating fan speed, or zero to indicate that the fan is off. There MAY be a temporary mismatch between the value of this attribute and the value of the [PercentSetting](#) attribute due to other system requirements or constraints that would not allow the fan to operate at the requested setting.

For example, if the value of this attribute is currently 50%, and the [PercentSetting](#) attribute is newly set to 25%, the value of this attribute MAY stay above 25% for a period necessary to dissipate internal heat, maintain product operational safety, etc.

When the value of the [FanMode](#) attribute is AUTO, the value of this attribute MAY vary across the range over time.

See [Section 4.4.6.3.1, “Percent Rules”](#) for more details.

4.4.6.5. SpeedMax Attribute

This attribute SHALL indicate the maximum value to which the [SpeedSetting](#) attribute can be set.

4.4.6.6. SpeedSetting Attribute

This attribute SHALL indicate the speed setting for the fan. This attribute MAY be written by a client to indicate a new fan speed. If the [FanMode](#) attribute is set to Auto, the value of this attribute SHALL be set to null.

The server SHALL support all values between 0 and [SpeedMax](#).

If a client writes null to this attribute, the attribute value SHALL NOT change. If the fan is in a state where this attribute cannot be changed to the requested value, the server SHALL return INVALID_IN_STATE.

When this attribute is successfully written to, the server SHALL set the value of the [FanMode](#) and [PercentSetting](#) attributes to values that abide by the mapping requirements listed below.

4.4.6.6.1. Speed Rules

It is up to the server implementation to map between ranges of the [SpeedSetting](#) attribute, the [PercentSetting](#) attribute, and the [FanMode](#) attribute enumerated values that are supported as indicated in the [FanModeSequence](#) attribute per the requirements below.

There SHALL be a mapping between the values of the [FanMode](#) attribute and the values of this attribute by dividing the domain of values of this attribute into ranges as follows:

- The ranges SHALL be non-overlapping, contiguous, and in ascending order.